

Stat 651 Statistical Theory I, Fall 2025
Homework 1, due Friday, August 29, 11:59p.m.

Please answer the following questions and submit them through Gradescope.

1. Use R to simulate the following; state the values it generates. (Please ask me if you haven't used R before. It takes (maybe) 10 minutes to install R (for free!), and less than two minutes to run the R code below.)

(a) Five draws from the set $\{1, 2, 3, \dots, 10\}$ with replacement. R code:

```
sample( 1:10, 5, replace=TRUE )
```

(b) Five independent draws from a normal distribution with mean 1.0 and standard deviation 0.5. R code:

```
rnorm( 5, mean=1.0, sd=0.5 )
```

2. Evaluate the following integrals, by hand and also using Wolfram Alpha. (Please ask me if you are unfamiliar with Wolfram Alpha. It's free online software; use google or duckduckgo to find it.)

$$I_1 = \int_0^1 x^2 dx =$$

$$I_2 = \int_0^1 \int_0^{2y} x dx dy =$$

Please sketch the region that I_2 corresponds to. (This is a Calculus III question, and Calc III is a prerequisite for this course; you may need a refresher on this when we evaluate these types of integrals halfway through the semester. Ask me if you have concerns about this.)

3. For each of the following experiments, describe the sample space. (List the outcomes if there are fewer than 17.) How many outcomes are there for each – finite, countably infinite, or uncountably infinite?
 - (a) Toss four distinct coins once. (Penny, Nickel, Dime, Quarter)
 - (b) Toss four indistinguishable coins once.
 - (c) Count the number of leaves on a plant that are over 2" in length.
 - (d) Measure the lifetime (in hours) of a particular brand of AAA battery.
 - (e) Record the shoulder heights of newborn foals.
 - (f) Observe the proportion of defective light bulbs in packs of 50 bulbs.